# VISVESVARAYA TECHNOLOGICAL UNIVERSITY

**'JNANA SANGAMA' BELAGAVI-590 018, KARNATAKA**

**PROJECT REPORT**

**ON**

**"Blockchain-Based Identity Verification for IT companies"**

**SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENT FOR THE AWARD OF THE DEGREE,**

**BACHELOR OF ENGINEERING**
**IN**
**COMPUTER SCIENCE & ENGINEERING**

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**2025-26**

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**2025-26**

**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING**

## CERTIFICATE

This is to certify that the project work entitled **"Blockchain-Based Identity Verification for IT companies"** has been successfully carried out by **Amith A N [1CG22CS008], Mithun M [1CG22CS065], Jayasurya N [1CG22CS047], Paartha K [1CG22CS080],**bonafide students of **CHANNABASAVESHWARA INSTITUTE OF TECHNOLOGY, GUBBI, TUMAKURU,** under our supervision and guidance and submitted in partial fulfillment of the requirements for the award of Degree in **Bachelor of Engineering** by **Visvesvaraya Technological University, Belagavi** during the academic year of 2025–2026. It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report deposited in the departmental library. The project report has been approved as it satisfies the academic requirements for the above said degree.

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# UNDERTAKING

We the students **Amith A N [1CG22CS008], Mithun M [1CG22CS065], Jayasurya N [1CG22CS047], Paartha K [1CG22CS080]** of **VII semester B.E. Computer Science and Engineering** of **CHANNABASAVESHWARA INSTITUTE OF TECHNOLOGY, GUBBI, TUMAKURU** declare that Project work entitled **"Blockchain-Based Identity Verification for IT companies"** has been carried out and submitted in partial fulfillment of the requirements for the award of degree in Bachelor of Engineering in **Computer Science and Engineering** by the Visvesvaraya Technological University during the academic year 2025-2026.

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**2025-26**

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# BONAFIDE CERTIFICATE

This is to certify that the Project work entitled **"Blockchain-Based Identity Verification for IT companies"** is a bonafide work of **Amith A N [1CG22CS008], Mithun M [1CG22CS065], Jayasurya N [1CG22CS047], Paartha K[1CG22CS080]** students of **VII** semester B.E computer Science and Engineering carried out at **Channabasaveshwara Institute of Technology**, Gubbi, Tumakuru, in partial fulfillment of the requirements of the award of degree in B.E. in **Computer Science and Engineering** of Visvesvaraya Technological University, Belagavi under my supervision and guidance. Certified that to the best of my knowledge the work reported here in does not from part of any other thesis on the basis of which degree or award was conferred on earlier occasion to this or any other candidates.

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# ACKNOWLEDGEMENT

A great deal of time and lot of effort has gone into completing this project report and documenting it. The number of hours spent in getting through various books and other materials related to this topic chosen by us have reaffirmed its power and utility in doing this project.

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We wish to express our deep sense of gratitude to our guide to **Dr. Shantala C P** Professor & Head of the Department of Computer Science, for her meticulous attention to details, which has contributed immeasurably to the quality of the project report

We would express our gratitude towards our parents and friends for their kind cooperation and encouragement which helped us in completion of this project.

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Thanking everyone....

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## ABSTRACT

The increasing need for secure and transparent identity verification systems has driven the adoption of blockchain technology in managing digital identities. Traditional identity management systems often rely on centralized databases, which are prone to data breaches, unauthorized access, and manipulation. To overcome these limitations, this project introduces a Blockchain-Based Employee Profile Verification System, designed to ensure data integrity, privacy, and trust among employees, universities, and organizations.

The proposed system leverages Ethereum blockchain and smart contracts written in Solidity to store and verify employee profiles in a decentralized manner. Each employee's educational qualifications, work experience, and documents are securely recorded on the blockchain, eliminating the need for third-party intermediaries. The use of Metamask wallet allows users to interact directly with the blockchain through a web-based interface, while the Sepolia Test Network serves as a secure testing environment before mainnet deployment. Additionally, tools such as Remix IDE for contract development and SHA-256 hashing for data integrity further enhance the system's robustness.

The project ensures that universities can validate academic credentials, companies can verify employment history, and employees can maintain full control over their digital identity. By integrating decentralized verification, the system minimizes fraudulent activities and promotes trust between stakeholders. In the long term, this framework can be extended to support cross-industry identity management, integrating technologies like IPFS for document storage and AI-driven verification for improved accuracy. Overall, the project demonstrates how blockchain can revolutionize identity verification, making it tamper-proof, transparent, and self-sovereign for a more secure digital ecosystem.

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# List of Keywords

| | | |
|---|---|---|
| Blockchain | Ethereum | Smart Contract |
| Decentralized | Employee Profile Management | Identity Verification |
| Solidity | Remix IDE | MetaMask |
| Sepolia Test Network | SHA-256 | Cryptography |
| Digital Identity | Immutability | Data Integrity |
| Transparency | Audit Trail | Document Verification |
| HR Module | University Module | Admin Module |
| Role-Based Access Control | Self-Sovereign Identity | IPFS |
| Google Cloud Web3 Faucet | Data Privacy | Distributed Ledger |
| Authentication | Access Control | Tamper-Proof |
| Automation | Scalability | Efficiency |

# CHAPTER 1

## INTRODUCTION

Blockchain technology has emerged as one of the most transformative innovations of recent times, offering a decentralized and transparent approach to storing and managing transactional data. In contrast to conventional centralized databases, it functions on a distributed ledger system that guarantees immutability, security, and reliability—eliminating the need for any central governing authority. These characteristics make blockchain particularly valuable in sectors where maintaining data accuracy and verification is essential

This project is centered on building an Employee Profile Management System powered by Blockchain technology, utilizing Ethereum smart contracts. The smart contract, developed in Solidity, was deployed on the Sepolia Test Network through Remix IDE, while MetaMask functioned as the connected wallet interface. For deployment and testing purposes, Sepolia test tokens were obtained from the Google Cloud Web3 faucet, enabling transaction execution without the need for actual cryptocurrency.

The smart contract is structured to securely manage employee information, including attributes such as name, age, designation, and address, while also establishing links between employees and universities registered on the blockchain. Upon deployment, the contract is initialized with predefined university addresses to ensure that the system starts with verified and trusted entities. It employs mappings to organize employee and university records, and triggers events whenever a new employee is registered—facilitating seamless monitoring and future integration with a user interface.

By integrating employee records on the Ethereum blockchain, this project demonstrates how decentralized technology can effectively replace conventional centralized systems with a transparent and tamper-resistant framework. Unlike traditional HR databases that are prone to unauthorized modifications or data manipulation, a blockchain-driven approach guarantees data integrity, transparency, and accountability across all transactions

In summary, this project offers hands-on experience in blockchain development, covering the process from writing Solidity smart contracts to deploying them on the Sepolia test network using practical tools. It highlights the capabilities of blockchain for modern record management and lays the groundwork for future enhancements, including the creation of a web-based interface, implementation of role-based access controls, and integration of secure document storage solutions.

## 1.1 PROBLEM STATEMENT

Current identity verification methods in IT recruitment are heavily centralized, prone to document forgery, time delays, and dependence on intermediaries, which compromises both efficiency and trust

## 1.2 OBJECTIVES

- To build a secure, decentralized Employee Profile Management System using blockchain
- To develop and deploy smart contracts on the Ethereum Sepolia Test Network
- To integrate trusted institutions for document and record verification
- To demonstrate how blockchain improves HR record-keeping

# CHAPTER 2

## LITERATURE SURVEY

1. **Title: Developing an IoT identity management system using blockchain Systems**

   **Author: Venkatraman, Sitalakshmi, and Sazia Parvin.**

   **Published Year:2022**

Identity (ID) management systems have evolved based on traditional data modelling and authentication protocols that are facing security, privacy, and trust challenges with the growth of Internet of Things (IoT). Research surveys reveal that blockchain technology offers special features of self-sovereign identity and cryptography that can be leveraged to address the issues of security breach and privacy leaks prevalent in existing ID management systems. Although research studies are recently exploring the suitability of blockchain based support to existing infrastructure, there is a lack of focus on IoT ecosystem in the secured ID management with data provenance of digital assets in businesses. In this paper, we propose a blockchain based ID management system for computing assets in an IoT ecosystem comprising of devices, software, users, and data operations. We design and develop a proof-of-concept prototype using a federated and distributed blockchain platform with smart contracts to support highly trusted data storage and secure authentication of IoT resources and operations within a business case scenario.

2. **Title: BLOCKCHAIN-BASED IDENTITY MANAGEMENT SYSTEMS**

   **Authors: Liu, Y., He, D., Obaidat, M. S., Kumar, N., Khan, M. K.**

   **Published Year:2020**

Identity management solutions are generally designed to facilitate the management of digital identities and operations such as authentication, and have been widely used in real-world applications. In recent years, there have been attempts to introduce blockchain-based identity management solutions, which allow the user to take over control of his/her own identity (i.e. self-sovereign identity). In this paper, we provide an in-depth review of existing blockchain-based identity management papers and patents published between May 2017 and January 2020. Based on the analysis of the literature, we identify potential research gaps and opportunities, which will hopefully help inform future research agenda

3. **Title: Decentralizing privacy Using blockchain to protect personal data**
   **Authors: Zyskind, G., Nathan, O., & Pentland**

   **Published Year:2015**

The recent increase in reported incidents of surveillance and security breaches compromising users' privacy call into question the current model, in which third-parties collect and control massive amounts of personal data. Bit coin has demonstrated in the financial space that trusted, auditable computing is possible using a decentralized network of peers accompanied by a public ledger. In this paper, we describe a decentralized personal data management system that ensures users own and control their data. We implement a protocol that turns a block chain into an automated access-control manager that does not require trust in a third party. Unlike Bit coin, transactions in our system are not strictly financial -- they are used to carry instructions, such as storing, querying and sharing data. Finally, we discuss possible future extensions to block chains that could harness them into a well-rounded solution for trusted computing problems in society.

4. **Title: Blockchain's roles in strengthening cybersecurity and protecting privacy**
   **Authors: Kshetri N**

   **Published Year: 2017**

Blockchain is touted as a technology that can possibly provide a robust and strong cybersecurity solution and high level of privacy protection (Schutzer, 2016). Its proponents argue that this technology is secure by design. In a blockchain model, there is no need to store information with third parties. The records are on many interlocked computers that hold identical information. If one computer's blockchain updates are breached, the system rejects it (Kestenbaum, 2017). In addition, multi-signature protection or the requirement of more than one key to authorize a transaction process can further improve security and privacy

5. **Title: A first look at identity management schemes on the blockchain**
   **Authors: Dunphy, Paul, and Fabien AP Petitcolas**
   **Published year:2018**

The emergence of distributed ledger technology (DLT) based on a blockchain data structure has given rise to new approaches to identity management that aim to upend dominant approaches to providing and consuming digital identities. These new approaches to identity management (IdM) propose to enhance de-centralization, transparency, and user control in transactions that involve identity information; however, given the historical challenge to design IdM, can these new DLT-based schemes deliver on their lofty goals? We introduce the emerging landscape of DLT-based IdM and evaluate three representative proposals—uPort, ShoCard, and Sovrine—using the analytic lens of a seminal framework that characterizes the nature of successful IdM schemes.

6. **Title: A survey on blockchain-based identity management and decentralized privacy**
   **Authors: Gilani, Komal, et al**
   **Published Year:2020**

In the digital revolution, even secure communication between individuals, services and devices through centralized digital entities presents considerable risks. Having service providers continue to offer their centric solutions is inefficient in terms of duplication, has serious security lacunae and is cumbersome to the users. The self-sovereign Identity concept, which includes the individual's consolidated digital identity and verified attributes, enables the users of data to exert their ownership and gain insights from their data's usage. The authentication and verification of digital identity is essential to achieve the privacy and security of distributed digital identities. In this paper, we provide a coherent view of the central concepts of self-sovereign Identity, including the components of identity proofing and authentication solutions for different self-sovereign Identity solutions. We discussed an overview of Identity management approaches, introducing an architecture overview as well as the relevant actors in such a system and blockchain technology as solution for distributed user-centric identity.

7. **Title: Identity management systems for the IOT'S**
   **Authors: Zhu, Xiaoyang, and Youakim Badr.**
   **Published Year:2018**

The Internet of Things aims at connecting everything, ranging from individuals, organizations, and companies to things in the physical and virtual world. The digital identity has always been considered as the keystone for all online services and the foundation for building security mechanisms such as authentication and authorization. However, the current literature still lacks a comprehensive study on digital identity management for the Internet of Things (IoT). In this paper, we firstly identify the requirements of building identity management systems for IoT, which comprise scalability, interoperability, mobility, security, and privacy. Then, we trace the identity problem back to its origin in philosophy, analyze Internet digital identity management solutions in the context of IoT, and investigate recent surging blockchain sovereign identity solutions. Finally, we point out promising future research trends in building IoT identity management systems and elaborate on the challenges of constructing a complete identity management system for IoT, including access control, privacy preservation, trust, and performance, respectively.

8. **Title: Analysis of identity management systems using blockchain technology**
   **Authors: El Haddouti, Samia, and M. Dafir Ech-Cherif El Kettani.**
   **Published Year:2019**

The emergence of Blockchain technology, as one of the biggest innovations of the 21st century, has given rise to new concepts of Identity Management. These concepts aim to address privacy and security challenges on the one hand, and to enhance decentralization and user control in transactions on Blockchain infrastructures on the other hand. This paper investigates and provides an analysis of the most popular Identity Management Systems using Blockchain: uPort, Sovrin, and ShoCard. It then evaluates them under a set of digital identity features that characterize the success of an Identity Management solution. The results of the comparative analysis are presented in a concise manner to allow readers to easily identify which systems Satisfy specific requirements, enabling them to select the most appropriate solution for a

given scenario

9. **Title: Blockchain based Identity Management System**

   **Authors: C. Victoria Priscilla**

   **Published Year:2021**

Reliable identity management system is an integral part of the enterprise. Blockchain is a distributed ledger that is decentralized, cryptographically signed, with the ability to solve the protection and privacy problems of Centralized Identity Management System. The distributed consensus design of the blockchain will prevent malicious attacks, and enables self-sovereign identity by providing an individual to have more control over their information and to share minimum amount of information that are required for identification. This paper focused on the leveraging power of Blockchain technology in Digital Identity Management.

10. **Title: Blockchain aware decentralized identity management and access control system**

    **Authors: Aarti Amod Agarkar, Rajendra Talware, Gajanan Chavhan**

    **Published year:2024**

Identity Management Systems (IMS) manage individual profiles, user authorization, authentication, and access privileges. Paper-based profiles and defacement are vulnerable to spoofing, loss, and theft. Digital platforms with efficient IMS boost administrative processes and bring transparency and interoperability among establishments. The traditional paper-based identity systems are slow and rigid. Conventional centralized IMS lack transparency and are vulnerable to single-point failure. This paper presents a blockchain-aware decentralized identity man agement and access control (BADIMAC) model that mitigates these barriers. BADIMAC allows users to keep their digital identity within their control (i.e., self-sovereign identity). On the other hand, with the help of BADIMAC, service providers across enterprise boundaries can verify their identity without depending on any central au thority. BADIMAC offers transparency and better security. The decentralized consensus-based design of BAD IMAC enables the verifier to validate claims and establish proof of digital identity with process effectiveness.

# CHAPTER 3

## SYSTEM ANALYSIS

The current employee identity management systems rely on centralized databases, which are prone to data tampering, document forgery, and delays due to manual verification processes. These systems also offer limited transparency and are vulnerable to single points of failure, reducing trust and efficiency in HR operations. The proposed Blockchain-based Employee Profile Management System addresses these limitations by storing employee records on a decentralized Ethereum blockchain using smart contracts. This approach ensures data immutability, security, and verifiable audit trails while allowing automated document verification by HR and universities. The system employs role-based access controls and cryptographic techniques to protect sensitive information and provides transparency for all stakeholders. Using tools such as Solidity, Remix IDE, MetaMask, and the Sepolia Test Network, the system demonstrates a secure, efficient, and scalable method for managing employee identities, with potential for future enhancements like web-based interfaces and integrated document storage.

### 3.1 EXISTING SYSTEM

In traditional HR and identity management systems, employee data is stored in centralized databases maintained by organizations. These systems rely heavily on administrative control and third-party intermediaries to authenticate employee credentials. While functional, these approaches suffer from several drawbacks

- Data Manipulation Risks: Centralized databases are prone to unauthorized modifications, accidental deletions, or internal fraud.
- Document Forgery: Verification of educational qualifications, work experience, and other credentials is often manual, making the system vulnerable to forged documents.
- Time-Consuming Processes: Validation and approval procedures involve multiple intermediaries, leading to delays.

- Limited Transparency: Employees and external auditors have restricted access to verify records, reducing trust in the system.
- Single Point of Failure: If the central server or database fails or is attacked, the entire system becomes inaccessible, jeopardizing continuity.

**DISADVANTAGES**

- Vulnerability to Data Manipulation: Centralized databases can be altered or tampered with by unauthorized personnel.
- Document Forgery: Manual verification processes make it easier for fake documents to go undetected.
- Single Point of Failure: System crashes or cyberattacks can compromise the entire database.
- **Time-Consuming Verification**: Multi-step manual processes and reliance on intermediaries slow down employee onboarding and record verification.
- **Limited Transparency**: Employees and auditors have restricted visibility into the verification process, reducing trust.
- **Higher Operational Costs**: Continuous monitoring, validation, and administrative tasks increase operational overhead

## 3.2 PROPOSED SYSTEMS

The proposed Blockchain-based Employee Profile Management System leverages Ethereum smart contracts to store and manage employee records in a decentralized and tamper-proof manner. This system improves upon existing models through the following advantages

- **Decentralization**: Data is stored across multiple nodes on the blockchain, eliminating single points of failure.
- **Immutability**: Once added, records cannot be altered without consensus, ensuring authenticity and reliability

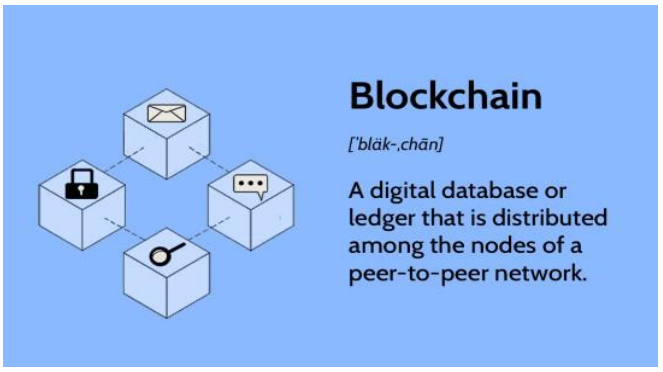
- **Automated Verification**: Smart contracts facilitate automated validation of documents through HR representatives and registered universities, reducing manual intervention.
- **Transparency and Auditability**: Employees, HR, and universities can track verification events and document status in real-time.
- **Enhanced Security**: Role-based access controls and cryptographic verification protect sensitive employee data from unauthorized access.
- **Efficiency**: By reducing dependency on intermediaries and manual processes, the system accelerates identity verification and employee onboarding workflows.

**ADVANTAGES**

- Decentralized Data Storage: Employee records are distributed across blockchain nodes, eliminating single points of failure.
- Data Immutability: Once recorded, information cannot be altered or deleted, ensuring authenticity and trustworthiness.
- Automated Verification: Smart contracts streamline validation of employee credentials by HR and universities without manual intervention.
- Enhanced Security: Cryptographic techniques and role-based access controls protect sensitive employee data from unauthorized access.
- Transparency and Auditability: All actions, such as document verification and updates, are recorded on the blockchain, providing a clear audit trail.
- Reduced Dependency on Intermediaries: The system minimizes reliance on third parties, speeding up verification and onboarding processes.
- Tamper-Proof Records: Blockchain ensures that all stored data is resistant to fraud, forgery, or unauthorized manipulation.
- Scalability for Future Enhancements: The system can be integrated with web interfaces, secure document storage, and role-based access controls in the future.

## 3.3 TOOLS AND TECHNOLOGIES

### 1. Blockchain




Blockchain is a decentralized and distributed ledger technology that securely records transactions across multiple nodes. It eliminates the need for a central authority by allowing all participants to verify and store data in a transparent and tamper-proof manner. Each transaction is stored in a block, linked chronologically to form a continuous chain, ensuring data integrity and trust. In this project, blockchain is used to provide secure, transparent, and verifiable management of information without relying on a centralized database.

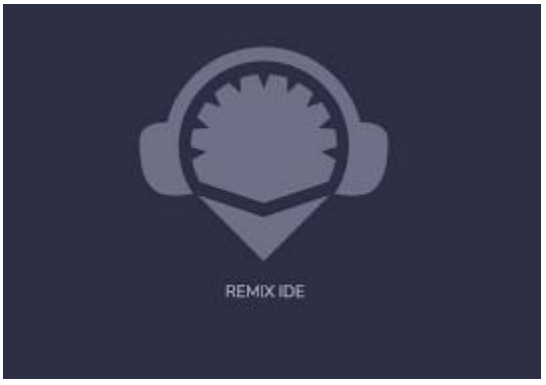
### 2. Ethereum Blockchain:

The core platform for deploying and running smart contracts. Ethereum provides a decentralized network that ensures immutability, transparency, and security of employee records. It eliminates the need for a central authority, making the system tamper-proof.

3. **Solidity:**

A high-level programming language specifically designed for writing smart contracts on the Ethereum blockchain. Solidity enables the creation of automated rules for employee record management, document verification, and role-based access control.

4. **Remix IDE:**

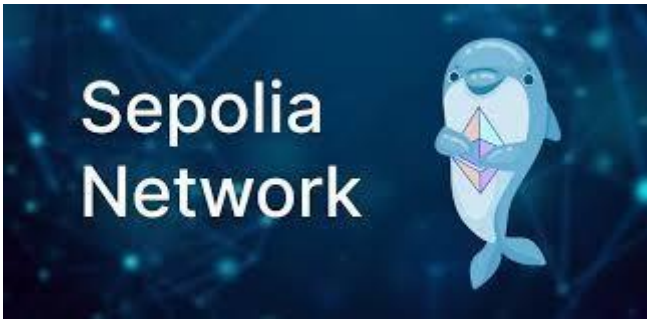



An online integrated development environment (IDE) used for writing, testing, and deploying Solidity smart contracts. Remix allows easy compilation, debugging, and deployment of contracts on Ethereum test networks like Sepolia**.**

5. **MetaMask:**

A browser-based cryptocurrency wallet that connects users to the Ethereum blockchain. MetaMask facilitates secure interaction with smart contracts, allowing employees, HR representatives, and universities to execute transactions without exposing private keys.

6. **Sepolia Test Network:**




A public Ethereum testnet used for deploying and testing smart contracts without spending real cryptocurrency. Sepolia provides a safe environment for testing transactions and verifying the system's functionality using test Ether

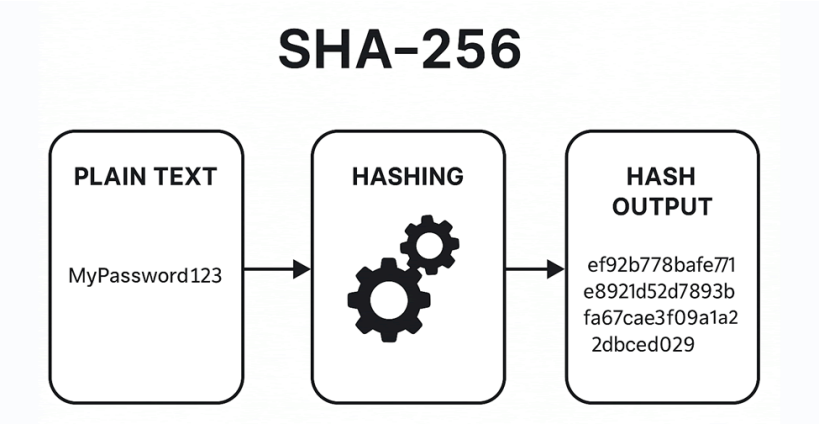
**7. Google Cloud Web3 Faucet:**

A source of Sepolia test tokens that allows users to conduct blockchain transactions during testing without the use of real Ether, enabling smooth development and experimentation.

**8. Smart Contracts:**

Self-executing contracts with coded rules for employee creation, document submission, verification, and profile management. They automate workflows, ensure data integrity, and provide a secure, auditable record of all actions.

**9. SHA256:**




SHA-256 is a widely used cryptographic hashing algorithm that ensures data integrity and security within blockchain systems. It converts any input data into a fixed 256-bit hash value, which acts as a unique digital fingerprint. Even a small change in the original data produces a completely different hash, making it highly resistant to tampering or forgery. In this project, SHA-256 is used to secure transaction data and employee records stored on the blockchain, ensuring that all information remains authentic, verifiable, and immutable.

## 3.4 System Requirements

### 3.4.1 Hardware Requirements:

- **i.** Processor: Intel Core i3 or higher (for Windows) / Apple M1 or higher (for macOS)
- **ii.** RAM: Minimum 8 GB for smooth execution of IDEs and browser-based tools
- **iii.** Storage: At least 256 GB HDD/SSD for installing development tools and storing project files
- **iv.** Graphics: Integrated graphics sufficient, no dedicated GPU required
- **v.** Internet Connection: Stable broadband connection for accessing blockchain networks, Remix IDE, and MetaMask

### 3.4.2 Software Requirements:

- **i.** Operating System: Windows 10/11 or macOS Monterey and above
- **ii.** Browser: Google Chrome or Mozilla Firefox (required for MetaMask extension)
- **iii.** MetaMask Wallet: Browser extension to interact with Ethereum blockchain and smart contracts
- **iv.** Remix IDE: Online platform for writing, compiling, and deploying Solidity smart contracts
- **v.** Solidity Compiler: Built-in in Remix IDE for compiling smart contracts
- **vi.** Node.js (Optional): For advanced testing or integration with web interfaces
- **vii.** Sepolia Test Network: Ethereum testnet used for deploying and testing smart contracts using test tokens

## 3.5 Functional Requirements

1. **Employee Profile Management**
   - Ability to create and store employee profiles with details such as name, age, role, and address on the blockchain,
   - Employees can update their educational qualifications and work experience
2. **Document Management**
   - Employees can upload multiple documents for verification,
   - HR and registered universities can verify documents through smart contracts**.**
3. **Role-Based Access Control**
   - Only authorized HR personnel or universities can verify employee documents.
   - Employees can view their own profiles but cannot modify verified documents.
4. **Automated Record Tracking**
   - Smart contracts emit events whenever a new employee or document is added, providing a transparent audit trail.
   - System maintains a tamper-proof log of verification and document status.
5. **Company and University Management**
   - Admin can register companies and universities on the blockchain
   - HR representatives can be assigned to respective companies.

## 3.6 Non - Functional Requirements

1. **Security**
   - Data stored on the blockchain is immutable and protected through cryptography, preventing unauthorized modifications.
2. **Scalability**
   - The system should handle multiple employees, companies, and universities without performance degradation
3. **Reliability**
   - Blockchain ensures continuous availability and resistance to system failures or attacks.
4. **Usability**
   - The interface should be intuitive, enabling employees, HR, and universities to interact with the system easily.
5. **Performance**
   - The system should provide quick response times for adding employees, documents, and verifying data.
6. **Transparency and Auditability**
   - All actions and transactions are recorded on the blockchain, allowing stakeholders to track changes and verifications in real-time.

# CHAPTER 4

# SYSTEM DESIGN

## 4.1 SYSTEM ARCHITECTURE

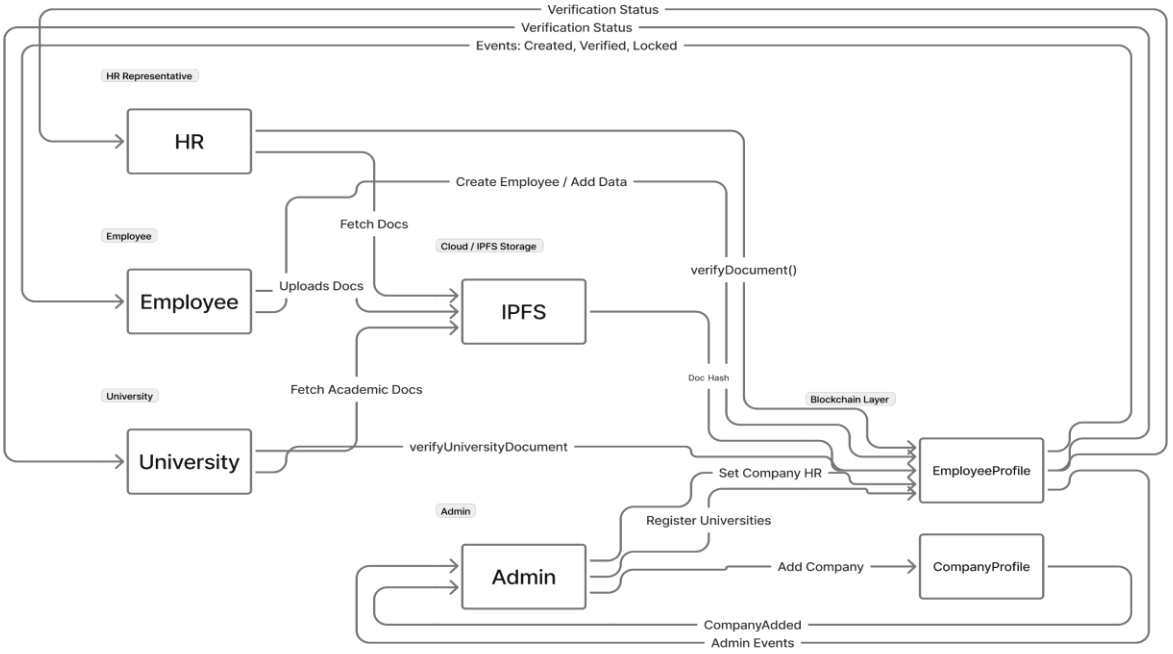



Fig 4.1 System Architecture

The system architecture of the Blockchain-based Employee Profile Management System is designed around a decentralized model using the Ethereum blockchain. It consists of multiple layers to ensure security, transparency, and efficiency. At the user layer, employees can create profiles, update their education and work experience, and upload documents for verification, while HR representatives and universities verify the submitted credentials. The application layer comprises smart contracts—EmployeeProfile and CompanyProfile—which handle all operations, enforce role-based access control, and emit events for tracking changes. At the blockchain layer, the Sepolia test network stores all employee, company, and verification data immutably, while MetaMask serves as a secure interface for users to interact with smart contracts. Operations such as profile creation, document submission, and verification are executed as blockchain transactions, ensuring tamper-proof, auditable records. Tools like Remix

IDE and Sepolia test tokens facilitate development, testing, and deployment, providing a reliable and decentralized framework for managing employee identities efficiently.

## 4.2 FLOWCHART:

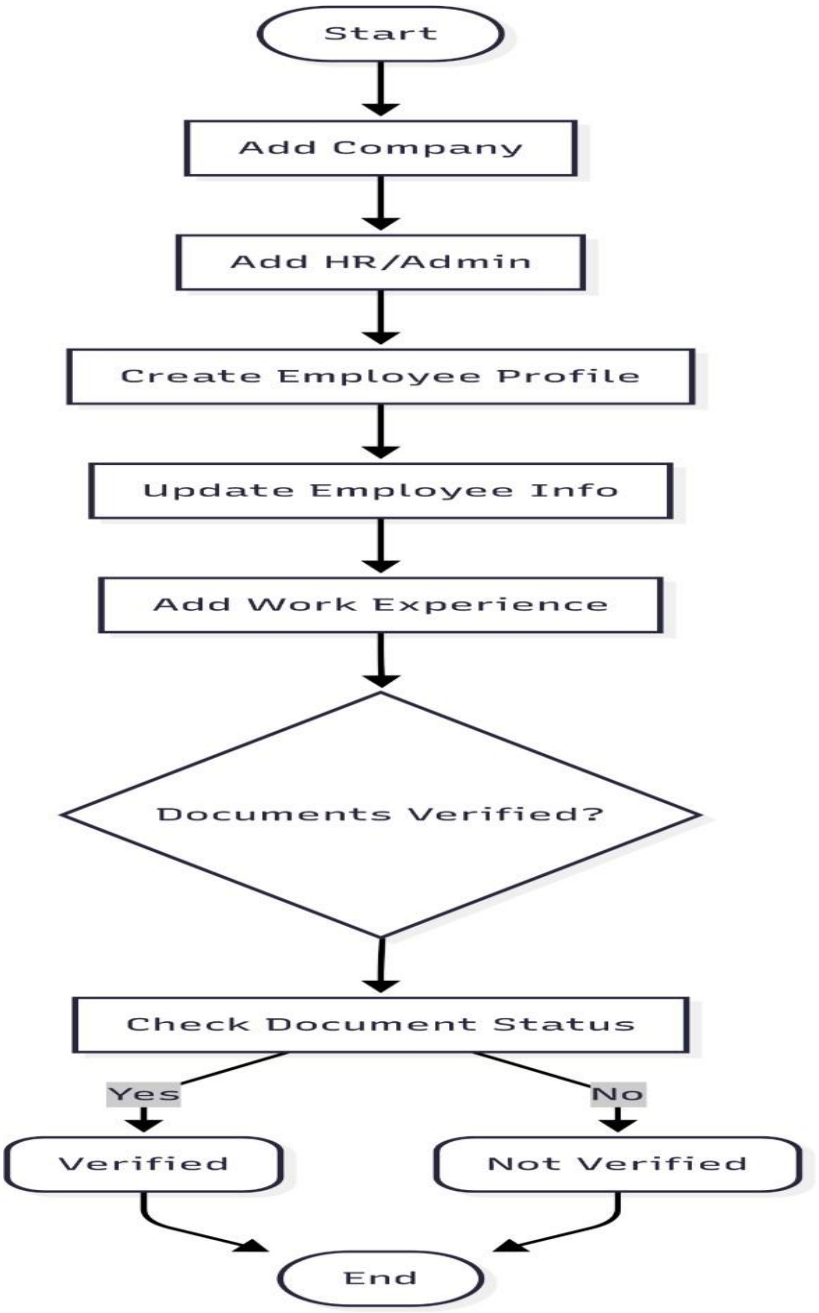



Fig 4.2 Flowchart

The flow of the system begins with the admin registering universities and companies on the blockchain using the setUniversity and addCompany functions. The admin also assigns HR representatives to companies via the setCompanyHR function.

Next, **employees create their profiles** by calling the createEmployee function, providing their personal details such as name and address. Once the profile is created, employees can **add educational qualifications and work experience** through the addEducation and addWorkExperience functions

For document management, employees use the addMultipleDocuments function to upload various documents, associating each document with the relevant company and university. After submission, the **verification process begins**: HR representatives verify documents using the verifyDocument function, while universities verify educational documents via the verifyUniversityDocument function. Both actions update the blockchain, marking documents as verified and locking the employee profile if necessary to prevent further edits.

Throughout this process, **events are emitted**—such as EmployeeCreated, EducationAdded, DocumentsAdded, DocumentVerified, and UniversityVerified—allowing stakeholders to track all actions transparently and maintain an immutable audit trail. The **MetaMask wallet** facilitates secure interactions between users and smart contracts, and the **Sepolia test network** ensures safe execution of all blockchain transactions without using real cryptocurrency.

This structured flow ensures that employee profiles are created, updated, and verified in a **secure, decentralized, and tamper-proof manner**, with clear accountability for every action taken by employees, HR personnel, or universities.

## 4.3 USECASE

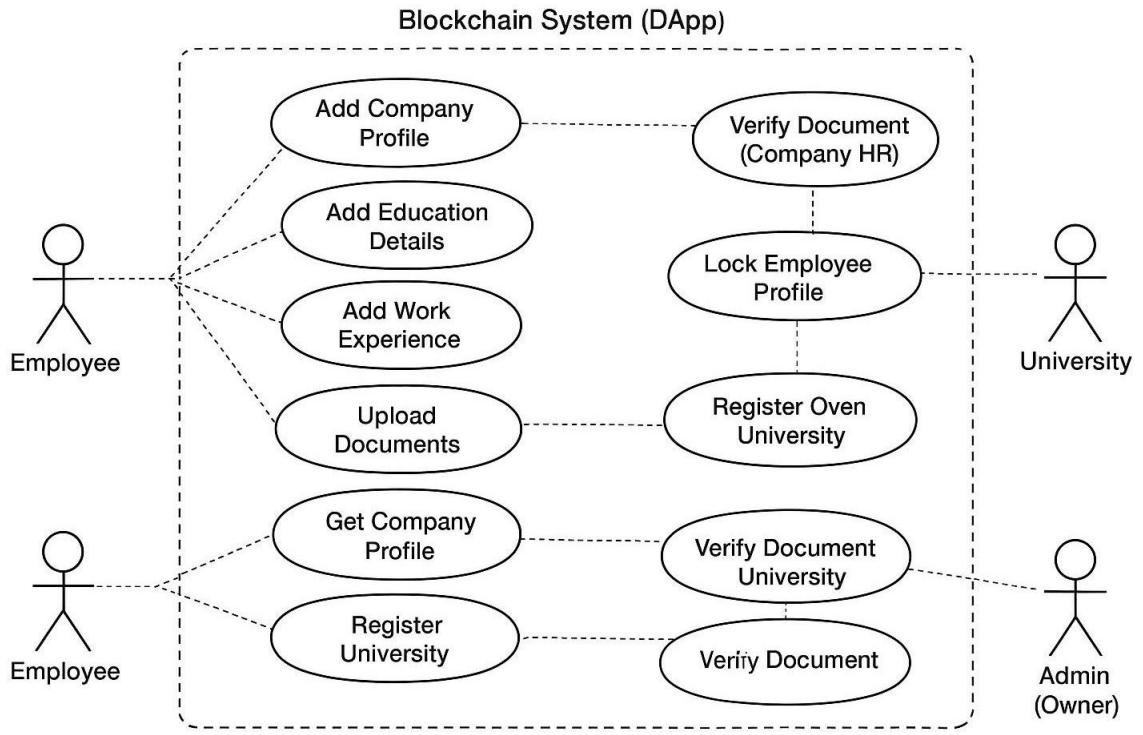



Fig 4.3 UseCase Diagram

## Actors:

- Employee
- HR Representative
- University
- Admin

1. **Create Employee Profile:**

- **Description:** An employee creates a personal profile by providing details such as name, address, and employee ID. The profile is stored securely on the Ethereum blockchain.

- **Actor:** Employee
- **Preconditions:** Employee has access to MetaMask and is connected to the Sepolia test network.
- **Postconditions:** Profile is successfully created, and an EmployeeCreated event is emitted.

2. **Add Education and Work Experience**

- **Description:** Employees can update their profiles with education qualifications and work experience.
- **Actor:** Employee
- **Preconditions:** Employee profile exists and is not locked.
- **Postconditions:** Information is stored on the blockchain, and events EducationAdded and WorkExperienceAdded are emitted.

3. **Upload Documents**

- **Description:** Employees upload supporting documents like certificates, associating them with companies and universities.
- **Actor:** Employee
- **Preconditions:** Employee profile exists.
- **Postconditions:** Documents are added to the employee profile, and a DocumentsAdded event is emitted.

4. **Verify Documents**

- **Description:** HR representatives or universities verify the uploaded documents. Verified documents trigger events and may lock the profile to prevent unauthorized edits.
- **Actors:** HR Representative, University
- **Preconditions:** Document is uploaded and associated with the HR/company or university.
- **Postconditions:** Document verification status is updated, and DocumentVerified or UniversityVerified events are emitted.

5. **Manage Companies and Universities**

- **Description:** Admin registers companies and universities on the blockchain and assigns HR representatives to respective companies.
- **Actor:** Admin
- **Preconditions:** Admin access is available.
- **Postconditions:** Companies and universities are registered, and HR roles are assigned; corresponding events are emitted.

6. **View Employee Data**

- **Description:** Authorized actors can view employee profiles, document statuses, and verification results.
- **Actors:** Employee, HR, University, Admin
- **Preconditions:** Profile exists.
- **Postconditions:** Data is retrieved from the blockchain securely and transparently.

# CHAPTER 5

## METHODOLOGY

The methodology for this project follows a structured approach to design, develop, and deploy a secure and transparent employee profile management system using blockchain technology.

1. **Requirement Analysis:**

   The first step involved a thorough study of the challenges present in conventional identity management systems. Traditional HR databases rely heavily on centralized servers, which makes them vulnerable to data tampering, unauthorized access, and single points of failure. Additionally, identity verification processes in IT recruitment are often slow and dependent on intermediaries, leading to delays and inefficiencies. The project team conducted a detailed analysis to identify both functional requirements—such as employee creation, document upload, and verification—and non-functional requirements, including security, reliability, scalability, and auditability. This step ensured that the proposed blockchain solution would directly address these existing limitations

2. **System Design:**

   After identifying the requirements, a decentralized system architecture was designed to leverage the benefits of blockchain technology. The system was divided into three main layers:

   - **User Layer:** Includes employees, HR representatives, universities, and admins. Employees can manage their profiles and upload documents; HR and universities are responsible for verifying documents; admins manage companies and assign HR roles
   - **Application Layer:** Consists of smart contracts written in Solidity. The EmployeeProfile contract manages employee details, education, work experience, and documents, while the CompanyProfile contract maintains company information and HR assignments. Role-based access control is enforced here to ensure that sensitive operations are performed only by authorized actors.

- **Application Layer:** Consists of smart contracts written in Solidity. The EmployeeProfile contract manages employee details, education, work experience, and documents, while the CompanyProfile contract maintains company information and HR assignments. Role-based access control is enforced here to ensure that sensitive operations are performed only by authorized actors.

Events were integrated into the smart contracts to provide an auditable record of all actions, such as profile creation, document upload, and verification. This design ensures accountability and allows all stakeholders to monitor changes in real time.

3. **Smart Contract Development:**

The development of smart contracts was a key step in this methodology. Using **Solidity**, the contracts were coded to handle all employee-related operations securely. Functions were implemented for creating employee profiles, updating education and work experience, uploading multiple documents, and verifying them by HR or universities. Special care was taken to incorporate modifiers that enforce role-based permissions and profile locking mechanisms, ensuring that once a document is verified, it cannot be tampered with. Additionally, event logging was embedded into each function to maintain transparency and traceability of all actions on the blockchain.

4. **Blockchain Integration and Testing:**

To deploy and test the smart contracts safely, the Sepolia Ethereum test network was used. Test tokens were obtained from the Google Cloud Web3 faucet and imported into **MetaMask**, which served as a secure interface for interacting with the blockchain. The contracts were written, compiled, and deployed using **Remix IDE**, providing an integrated environment for testing all functionalities. Extensive testing was conducted to validate the correct execution of functions, including employee creation, document upload, and verification workflows. Edge cases were considered, such as preventing duplicate employee IDs, ensuring correct role-based access, and validating document indices during verification.

5. **Deployment and Validation:**

After successful testing, the smart contracts were deployed on the Sepolia test network. Multiple test scenarios were executed to simulate real-world operations, such as employees creating profiles, submitting documents, and having them verified by HR and universities. The workflow demonstrated the system's ability to maintain **data integrity**, provide **transparent audit trails**, and ensure that only authorized stakeholders could modify or verify data. The use of blockchain eliminated reliance on centralized databases, significantly reducing the risk of data manipulation or loss.

6. **Security and Transparency Considerations**

A major part of the methodology focused on security and transparency. The decentralized nature of blockchain ensures that employee records are immutable and cannot be altered by unauthorized users. Cryptographic techniques inherent to Ethereum protect the data, while smart contracts enforce strict access control. The system also provides transparency: every action, from document submission to verification, emits an event that is publicly recorded on the blockchain, allowing stakeholders to audit activities in real time

7. **Future Enhancements:**

While the current implementation demonstrates the feasibility of blockchain-based employee profile management, the methodology also considers future enhancements. These include integrating a **web-based user interface** for easier interaction, implementing **role-based dashboards** for HR and universities, adding **secure document storage** off-chain with hash verification, and enabling **advanced analytics** on employee data. These improvements will make the system more scalable, user-friendly, and robust for enterprise use.

## 8. Summary

This methodology ensures a systematic and comprehensive approach to building a decentralized employee identity management system. By combining blockchain technology, Solidity smart contracts, MetaMask integration, and Sepolia testnet deployment, the project provides a secure, transparent, and tamper-proof solution for managing employee records. The structured workflow—from requirement analysis to testing and validation—demonstrates the practical application of blockchain for modern HR and identity management systems

This structured methodology ensures that the system is **secure, transparent, and efficient**, providing decentralized solution for employee identity management while minimizing risks associated with centralized databases.

CHAPTER 6

# Implementation

## Modules

The Blockchain-based Employee Profile Management System is implemented using a modular design to ensure maintainability, scalability, and security. Each module addresses a specific aspect of the system's functionality, and together they form a cohesive platform for managing employee identities in a decentralized manner. The modular approach also allows for future enhancements without affecting the entire system. The following modules have been implemented

1. **Employee Module**

   The Employee Module is designed to allow employees to create and manage their personal profiles on the blockchain securely. Employees can input details such as their name, employee ID, and Ethereum wallet address. Once registered, they can add their educational qualifications and work experiences, providing a complete overview of their professional background.

   One of the key features of this module is the ability to **upload multiple documents**, such as degree certificates, employment letters, or other credentials. Each document is associated with the corresponding company and university, ensuring proper verification workflows. Uploaded documents are linked to the employee profile in a secure, tamper-proof manner on the blockchain.

   This module empowers employees to **maintain ownership of their data**, a significant advantage over centralized systems. It ensures that employees have full control over their information while enabling verifiable interactions with HR representatives and universities.

2. **HR Module**

The HR Module handles the verification and management of employee profiles from a company perspective. Authorized HR representatives can access employee profiles submitted under their company and verify the uploaded documents. The verification process updates the document status on the blockchain, ensuring an **immutable record of all actions**.

Once verification is complete, the employee profile may be **locked** to prevent further modifications, ensuring that verified data remains consistent and trustworthy. HR representatives can also monitor employee activity and document submissions through events emitted by smart contracts, making auditing easier and transparent.

This module is crucial for maintaining **organizational trust** and ensuring that only legitimate employee records are approved for further processing, such as onboarding or promotion

3. **University Module**

The University Module is designed to verify academic credentials submitted by employees. Universities are registered on the blockchain by the admin and can approve or reject documents such as degree certificates or transcripts. When a university verifies a document, the blockchain updates the verification status, and the profile may be locked if necessary.

This module ensures **authenticity of educational qualifications** and protects against fraudulent claims. By using blockchain technology, universities can participate in a transparent and auditable system where their verification actions are permanently recorded, building trust among employers and employees alike.

4. **Admin Module**

The Admin Module provides overall control and management of the system. Admins are responsible for registering companies and universities, assigning HR representatives to their respective organizations, and monitoring system activity. This module ensures that only authorized entities interact with the system and provides a management interface for system

oversight.

Through the Admin Module, the system maintains its **structural integrity** by controlling access to sensitive actions and ensuring the correct mapping between employees, companies, and universities. Admins also have the ability to view all events and audit activities, which supports transparency and accountability.

5. **Document Management Module**

The Document Management Module acts as the backbone for handling all employee-related documents. Employees can submit multiple documents each tagged with the relevant company and university. HR representatives and universities can then verify these documents using dedicated smart contract functions.

Once documents are verified, events such as DocumentVerified or UniversityVerified are emitted, and employee profiles may be locked to prevent tampering. This module ensures that **document handling is secure, traceable, and immutable**, addressing the common challenges faced by traditional HR systems where documents are often stored in centralized databases susceptible to fraud or modification.

6. **Smart Contract Module**

The Smart Contract Module forms the core of the system, providing decentralized logic for all operations. It includes contracts like EmployeeProfile and CompanyProfile, which manage employees, companies, universities, and document verification processes**.**

Key functionalities include:

- Role-based access control to ensure only authorized actors can perform certain actions.
- Event emission to track all critical activities, such as profile creation, document addition, and verification.
- Immutable storage of all employee, company, and verification data on the blockchain, ensuring tamper-proof records.

The Smart Contract Module ensures security, transparency, and decentralization, making it the foundation of the system.

**7. Integration and Workflow**

All modules interact seamlessly to create a robust workflow. Employees submit their information and documents through the Employee Module. HR representatives and universities verify the submitted data via their respective modules. Admins manage the overall system, while the Smart Contract Module ensures secure execution of all operations on the blockchain.

This modular structure allows the system to scale easily, add new features, and adapt to additional requirements, such as a web-based interface, advanced analytics, or off-chain secure storage. By keeping modules separate but interconnected, the system achieves both flexibility and security, providing a decentralized solution for modern employee identity management.

**8. Sha256 Algorithim**

The **SHA-256 (Secure Hash Algorithm 256-bit)** is a cryptographic hash function widely used in blockchain systems to ensure data integrity, security, and immutability. It takes an input of any size—such as a transaction, block header, or digital identity record—and produces a fixed-length 256-bit (32-byte) hash value. SHA-256 is deterministic, meaning the same input will always generate the same output, yet it is computationally infeasible to reverse the process or derive the original input from the hash. Even a small change in the input results in a completely different hash value, a property known as the avalanche effect. In blockchain, SHA-256 is used to link blocks together by hashing the previous block's hash into the next block, making any tampering immediately detectable. It also plays a crucial role in authentication, digital signatures, and proof-of-work mechanisms, where miners must find a hash that satisfies specific difficulty constraints. Due to its strong resistance to collisions and preimage attacks, SHA-256 provides a high level of security and trust, making it suitable for blockchain-based identity management systems where data integrity, privacy, and verification are critical.

# CHAPTER 7

## RESULTS

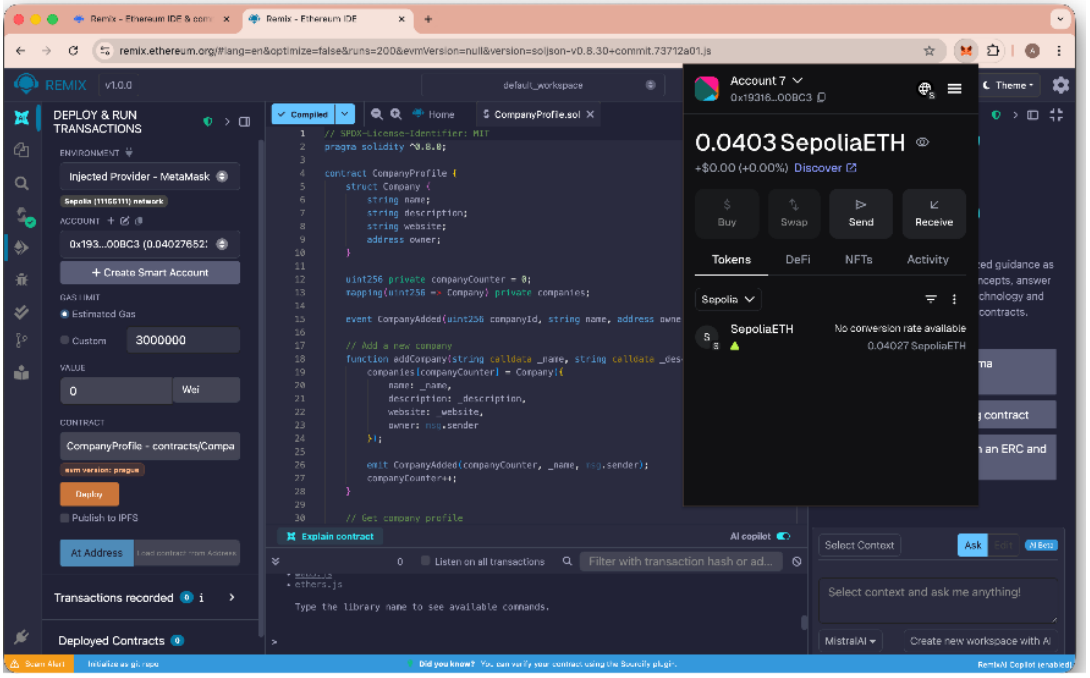



Fig 7.1 Successfully Got Sepolia free testtokens from Google Web3 Sepolia Faucet

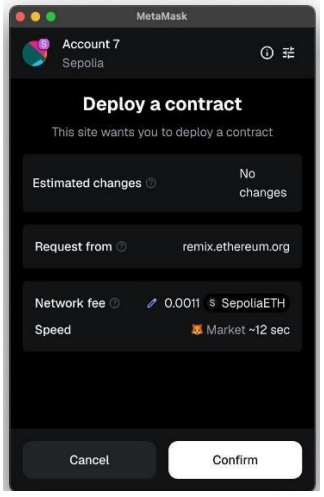



Fig 7.2 Deployed a contract in Remix ide by connecting Metamask to Remix

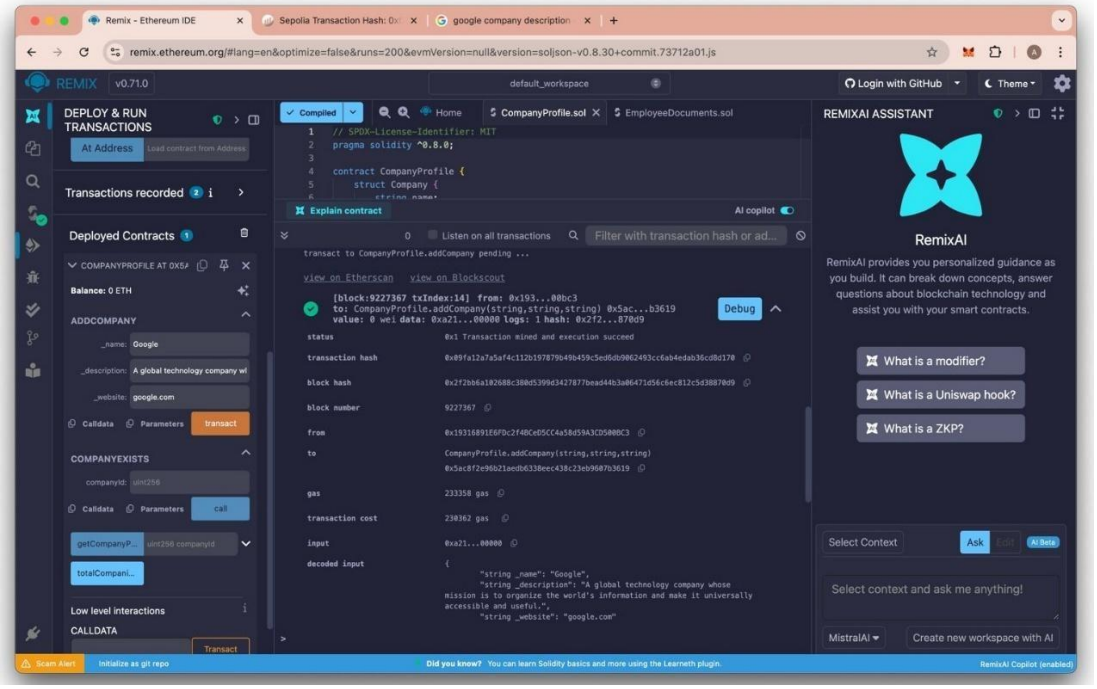



Fig 7.3 Successfully deployed and creation of Company Profile

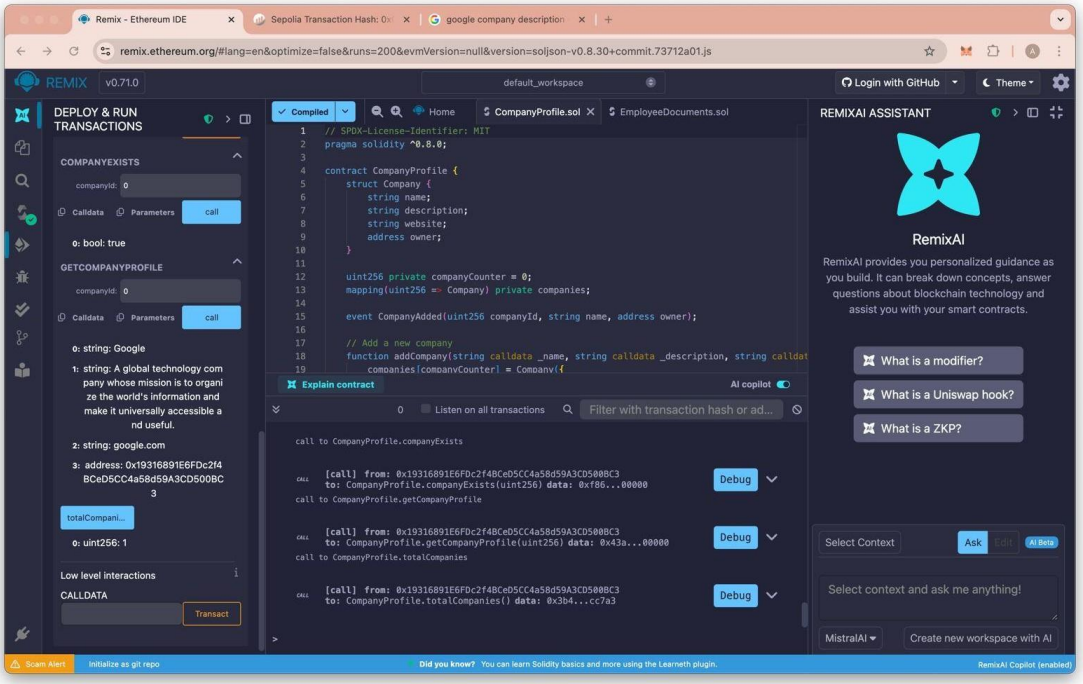



Fig 7.4 Retrieving the created Company Profile

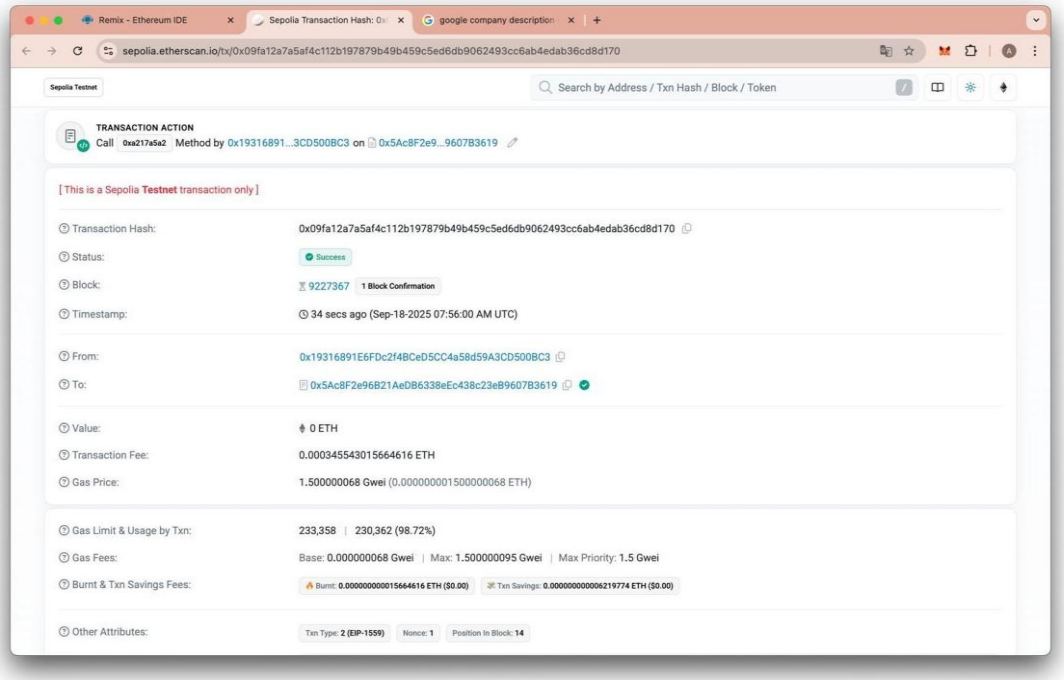



Fig 7.5 Confirmation of Block Creation of Company Profile Contract

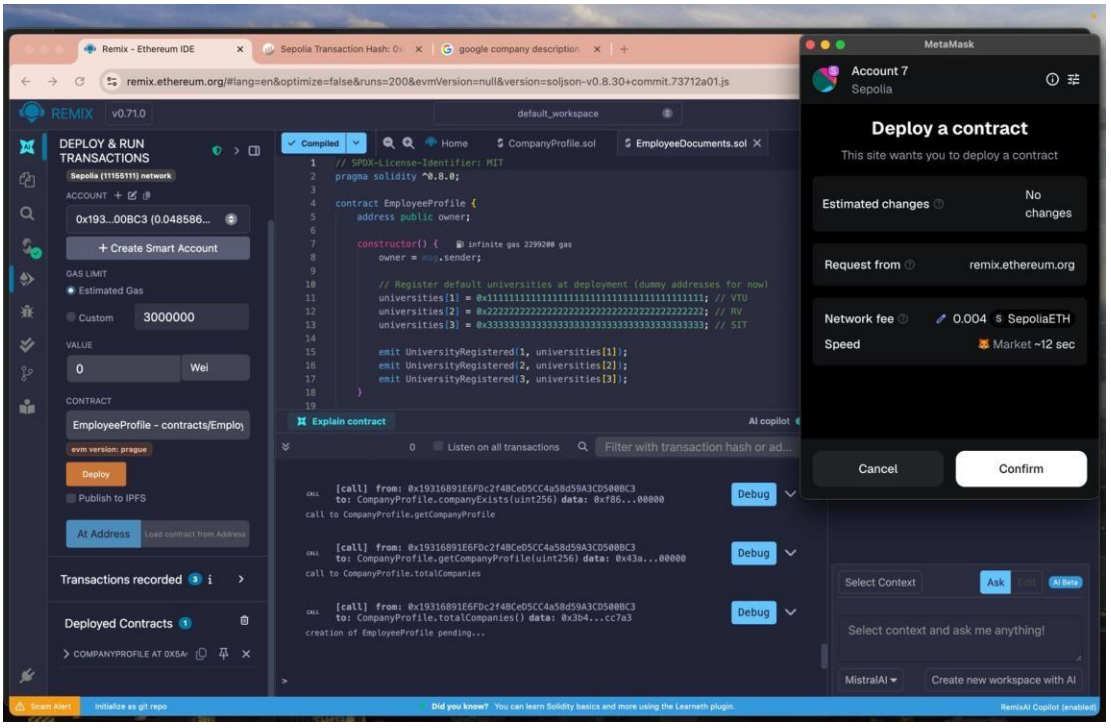



Fig 7.6 Requesting deployment of Employee Profile

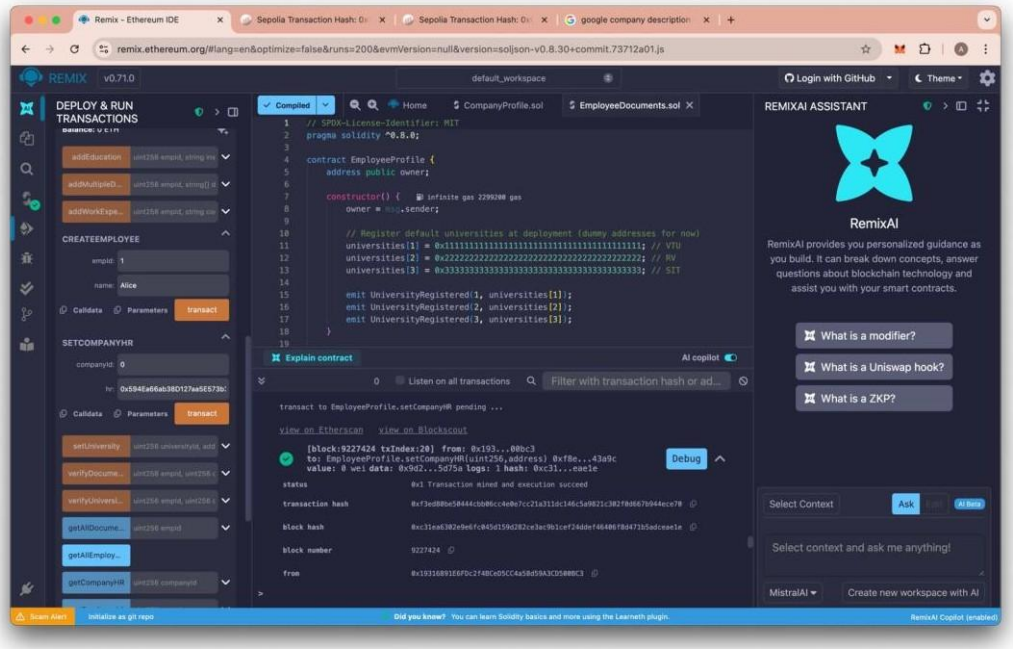



Fig 7.7 Successfully deployed and creation of Employee Profile (Creating Employee & Setting company HR)

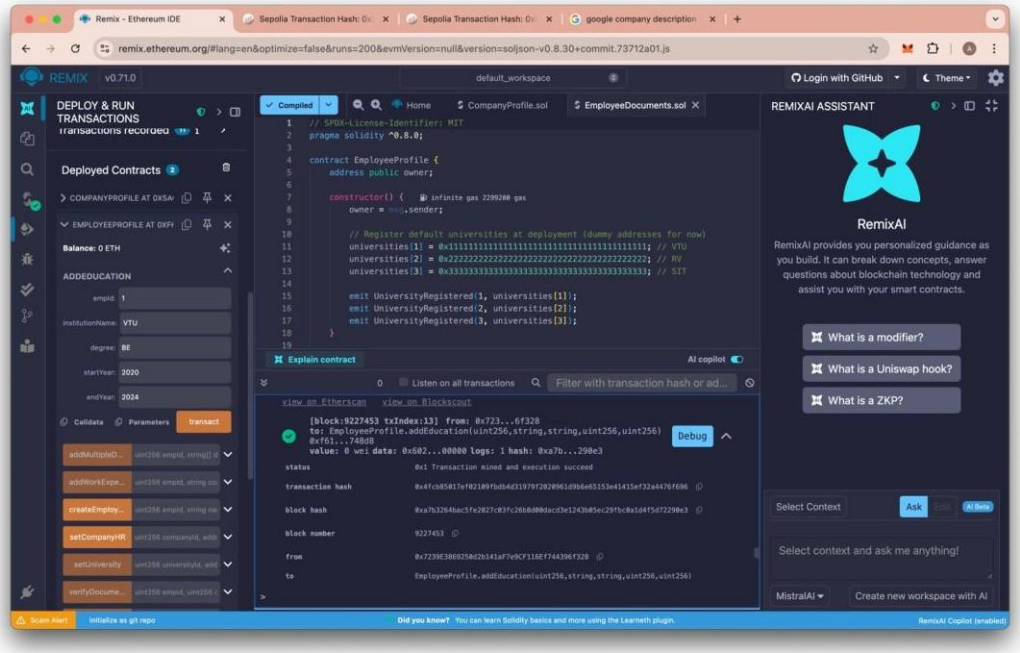



Fig 7.8 Successfully deployed and creation of Employee Profile (Adding Education Details)

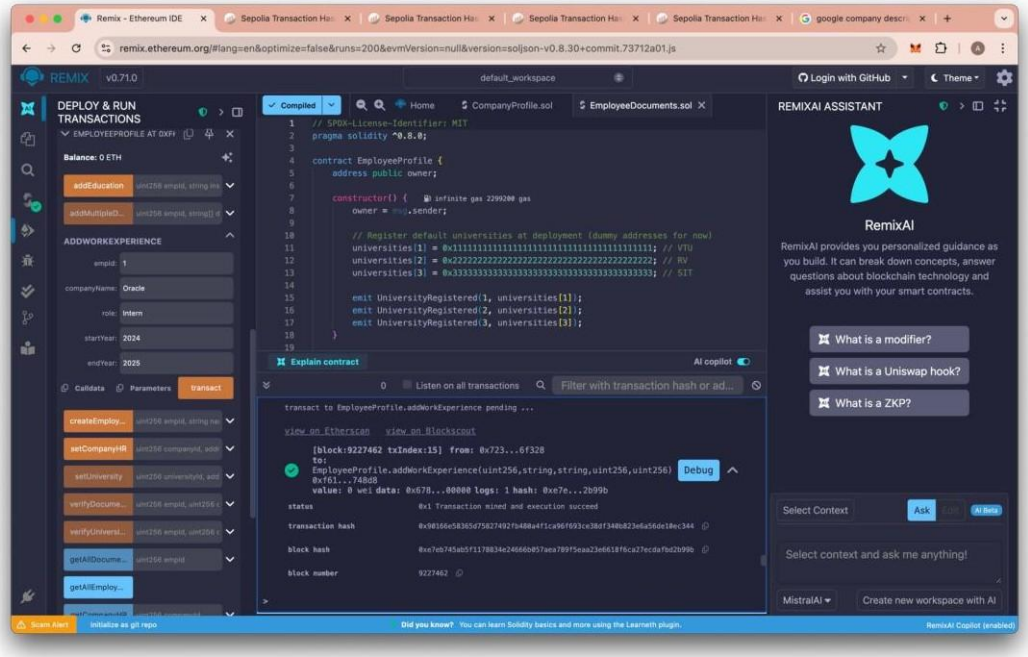



Fig 7.9 Successfully deployed and creation of Employee Profile (Adding Work Experience)

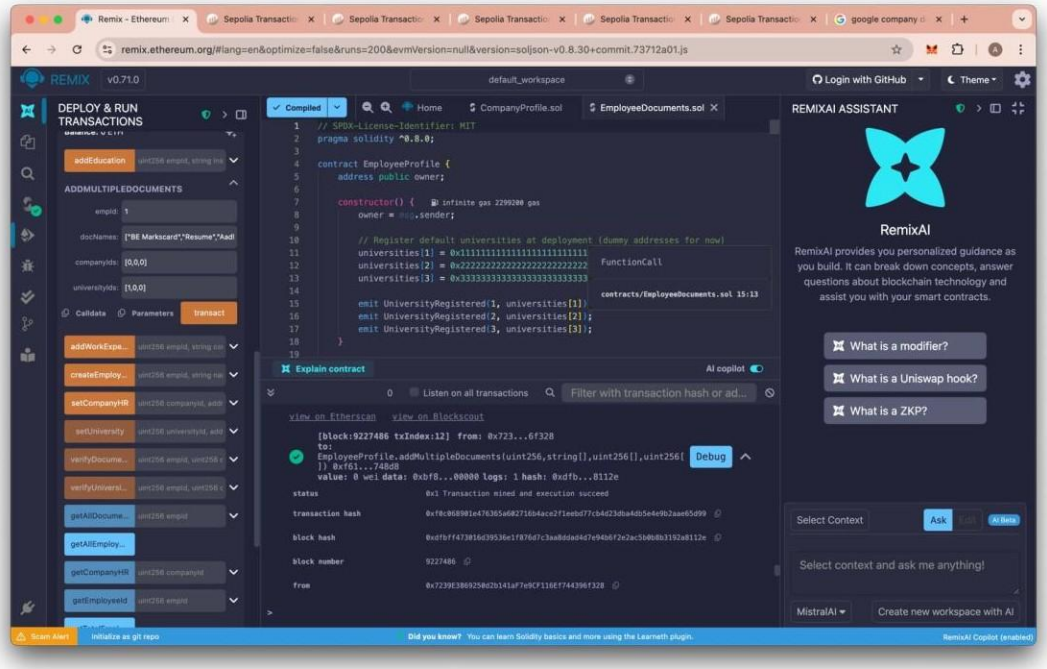



Fig 7.10 Successfully deployed and creation of Employee Profile (Adding Multiple Documents)

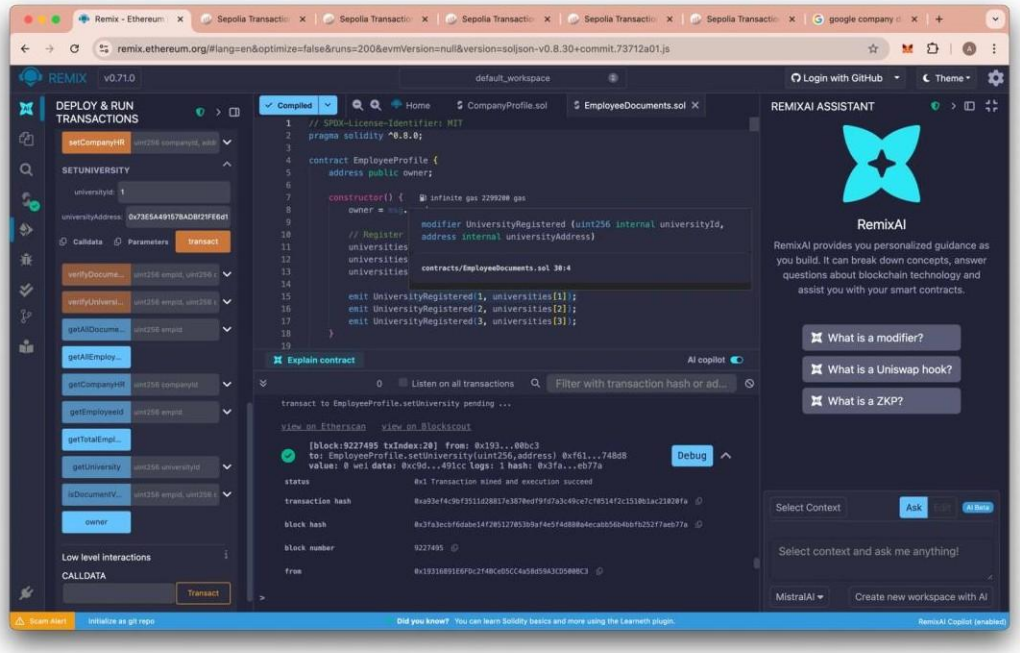



Fig 7.11 Adding Universities For Employee Profile

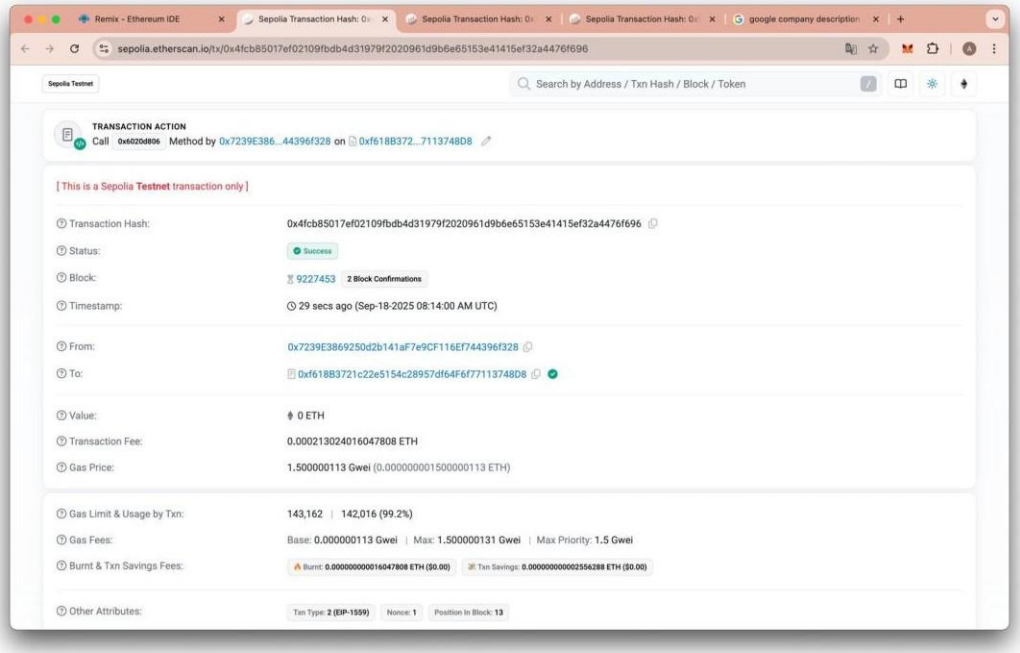



Fig 7.12 Confirmation of Block Creation of Company Profile Contract

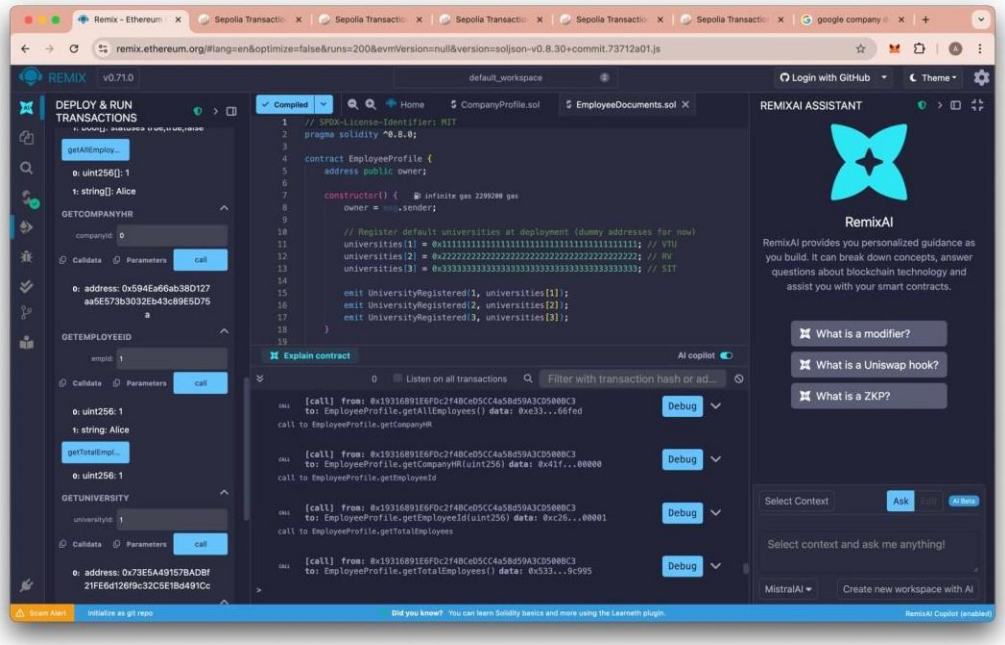



Fig 7.13 Retrieving the created Company Profile

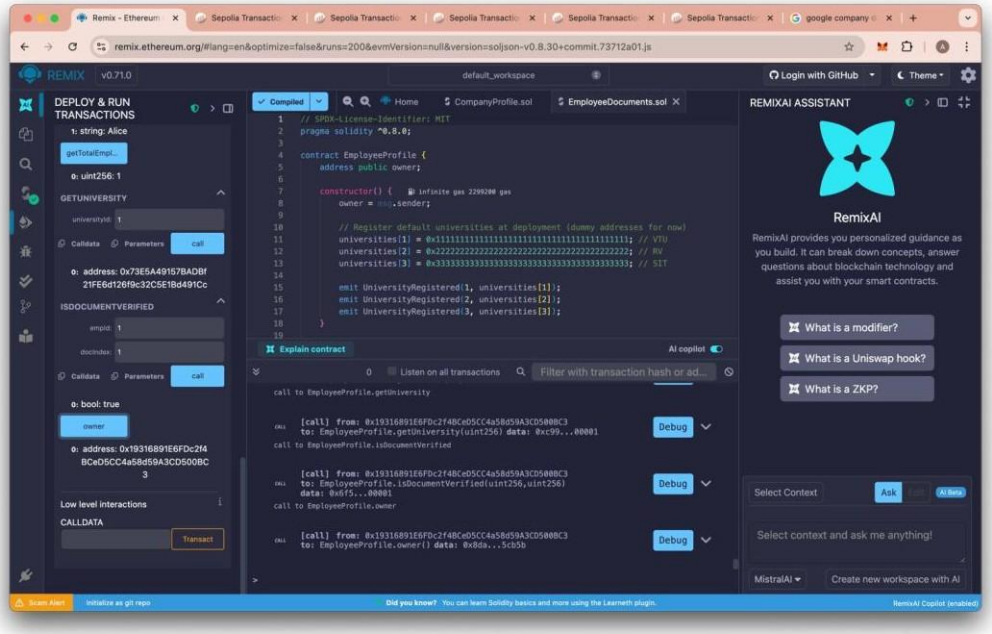



Fig 7.14 Retrieving the created Company Profile

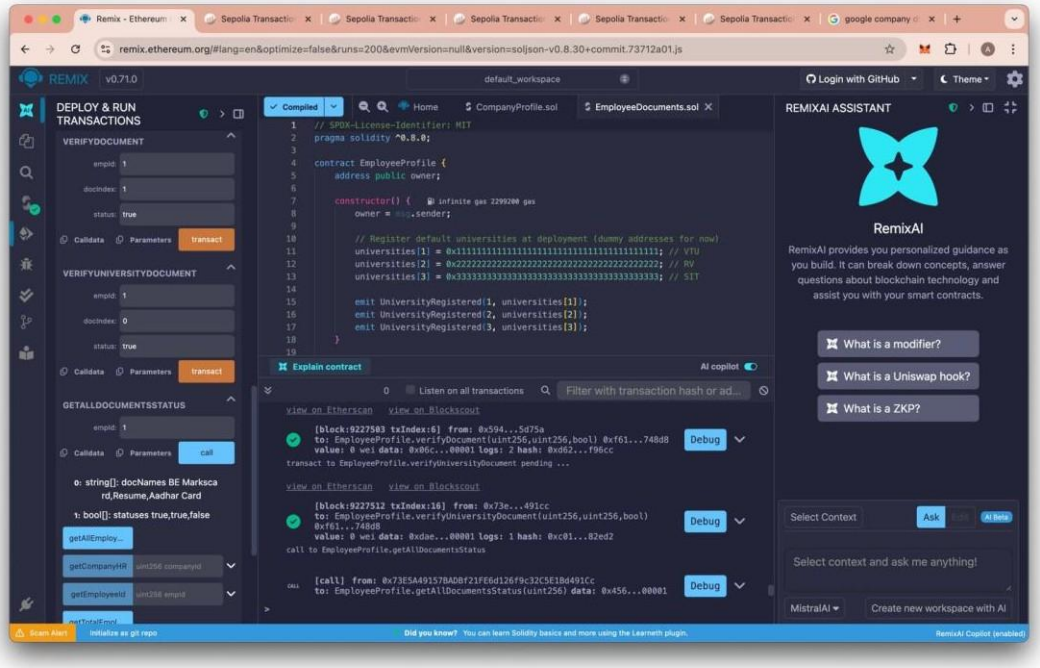



Fig 7.15 Verifying the Uploaded Doucments

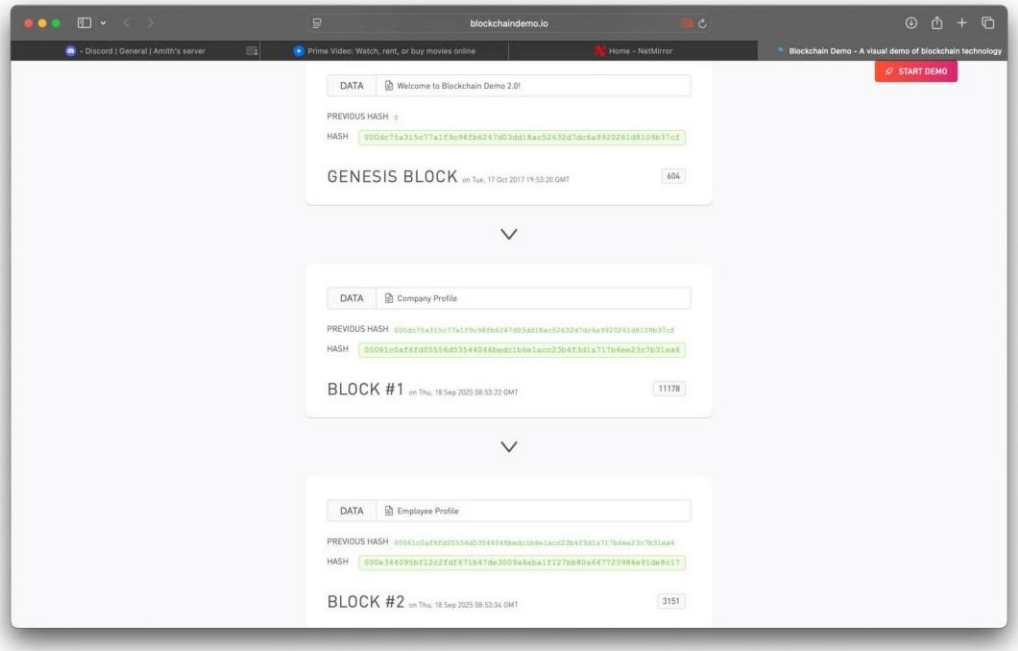



Fig 7.16 Showcasing the Creation of both CompanyProfile and EmployeeProfile

# CHAPTER 8

## CONCLUSION

The blockchain-based employee profile management system successfully demonstrates how decentralized technology can enhance transparency, security, and reliability in handling sensitive organizational data. By leveraging the immutable and tamper-proof nature of blockchain, this system ensures that employee information is protected against unauthorized access and manipulation. Unlike traditional centralized databases, the implementation provides a trustworthy platform where records are verifiable and auditable, fostering greater confidence among employees and management alike.

Moreover, the use of smart contracts automates key processes such as profile updates and access control, reducing manual errors and administrative overhead. The integration with test networks like Sepolia further validates the system's functionality in a real-world scenario, while also providing a safe environment for experimentation without affecting live data. Overall, this project highlights the practical benefits of blockchain in organizational settings and lays the foundation for future enhancements, such as a user-friendly web interface, expanded access permissions, and integration with other enterprise systems

## Future Enhancements

In the future, this blockchain-based employee identity management system can be enhanced by integrating an intuitive **web-based interface** where users, universities, and companies can interact seamlessly with the blockchain through a secure front end. Features such as **automated document verification using AI**, **QR code-based identity validation**, and **integration with decentralized storage systems** like IPFS can further strengthen the reliability and accessibility of the platform.

The system can also be extended to include **real-time analytics dashboards** for HR and universities to monitor verification status efficiently. These enhancements will not only improve user experience but also make the platform more secure, transparent, and adaptable for large-scale adoption in corporate and academic environments

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